

 **Digital Twin** 

Pub Crawl

Thursday,
20th of November

8.15pm onwards!



Meet us at...

Fox and Duck

The Bloomery

Notty House

Bar One

The Cavendish

Digital Twin



Phil Pöllmann

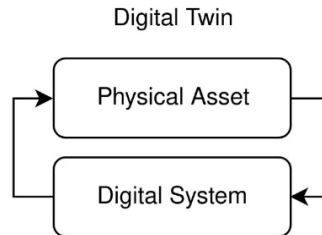


Jakob Titz



Digital Twins

- Virtual replica of a physical object, system or process
- Connected to the real world via data (e.g. sensors, logs, user inputs)
- Continuously updated to monitor, analyse and simulate behaviour
- Used to optimise performance, predict issues and support decisions





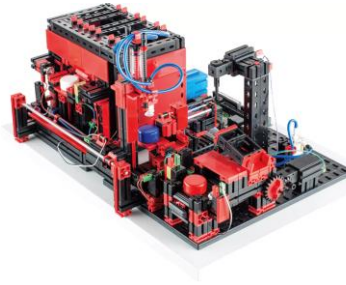
Our Physical Twin



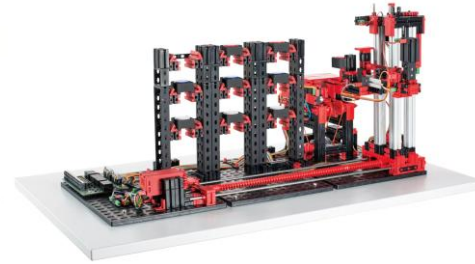


Setup

Multi Processing



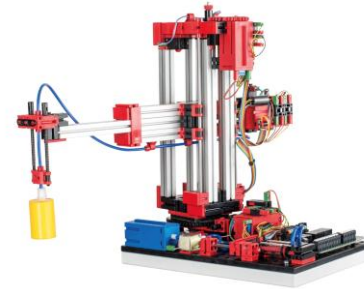
High Bay



Sorting Line



Conveyor Belt



Vacuum Gripper

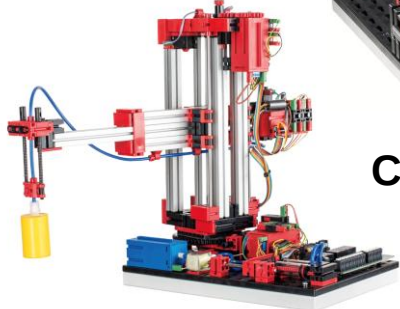


Setup

Sorting Line

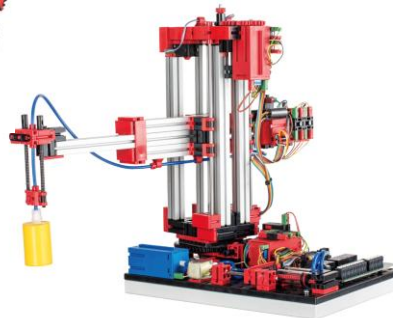
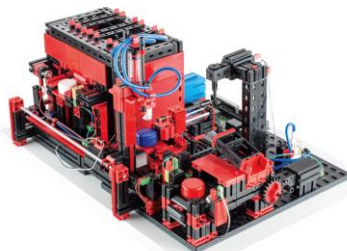


Conveyor Belt

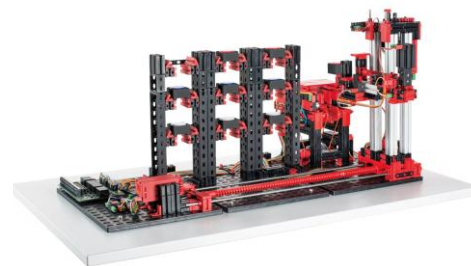


Vacuum Gripper 2

Multi Processing



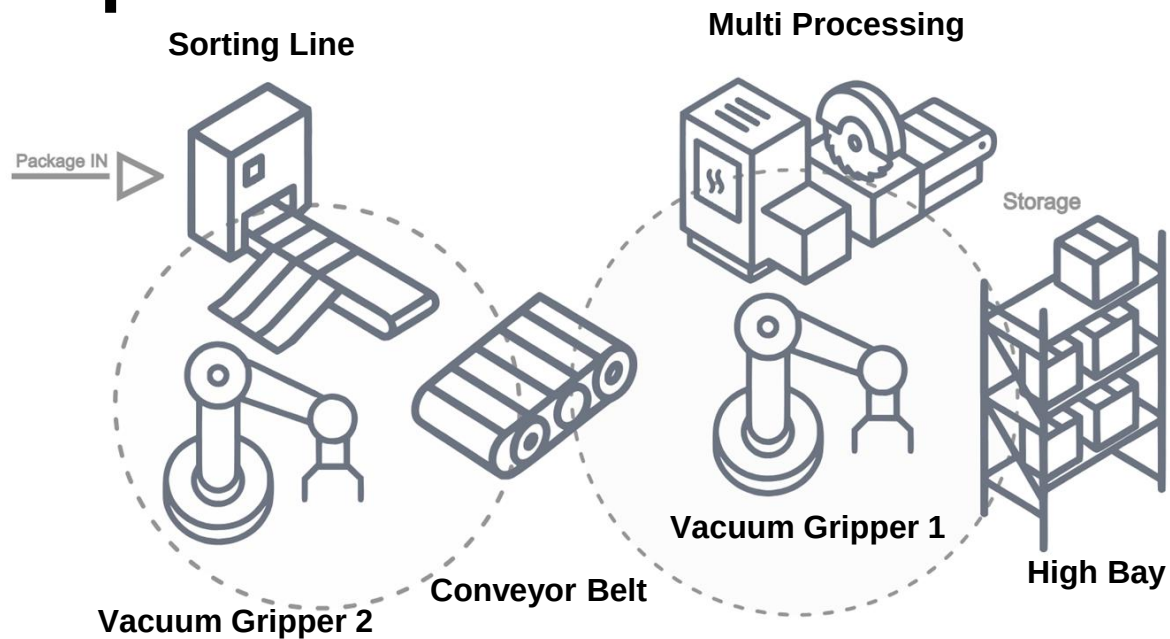
Vacuum Gripper 1



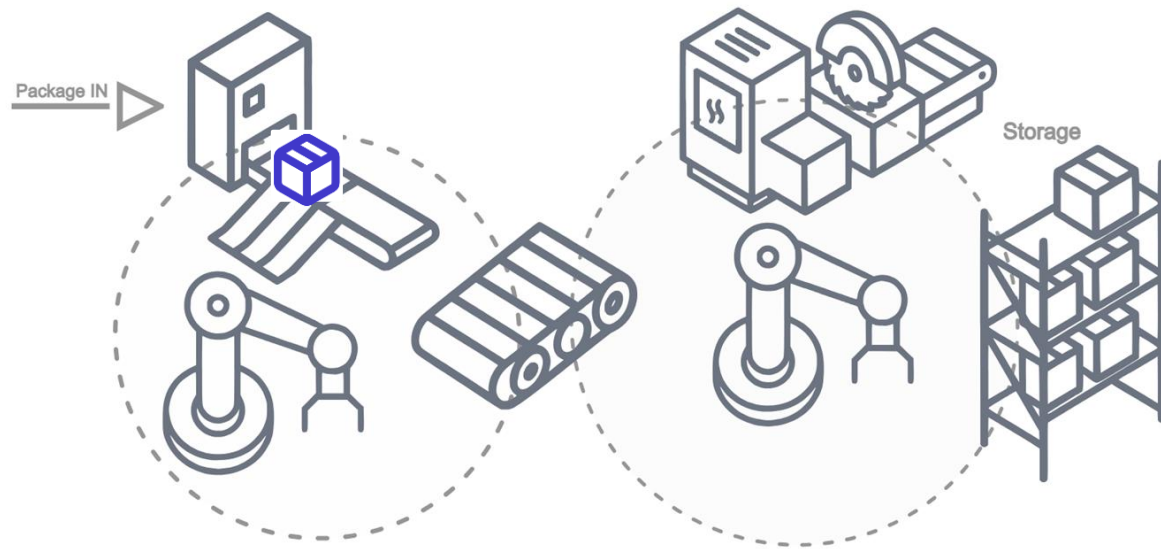
High Bay



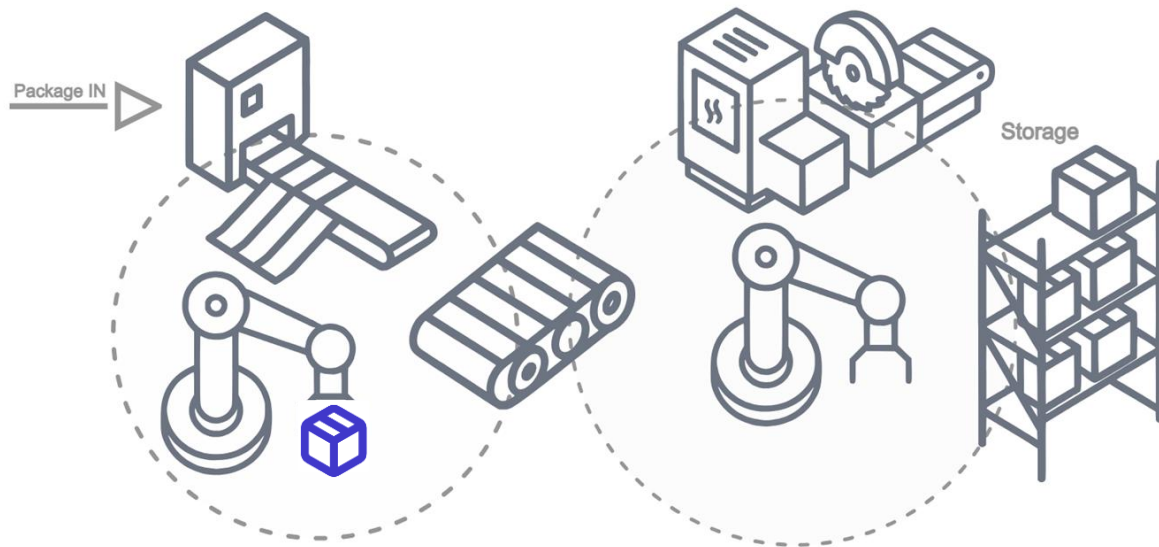
Setup



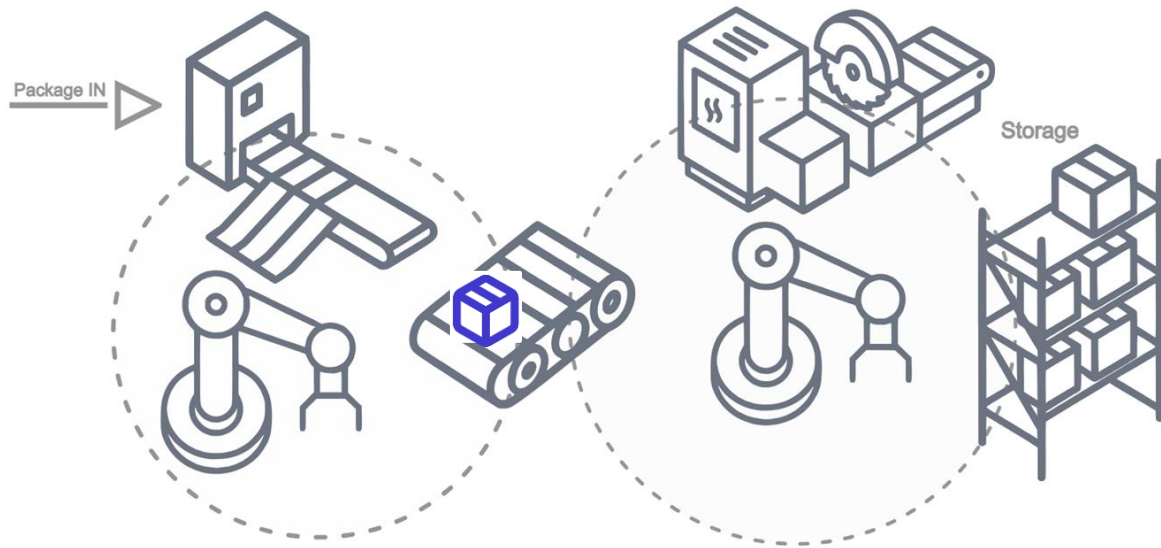
Setup



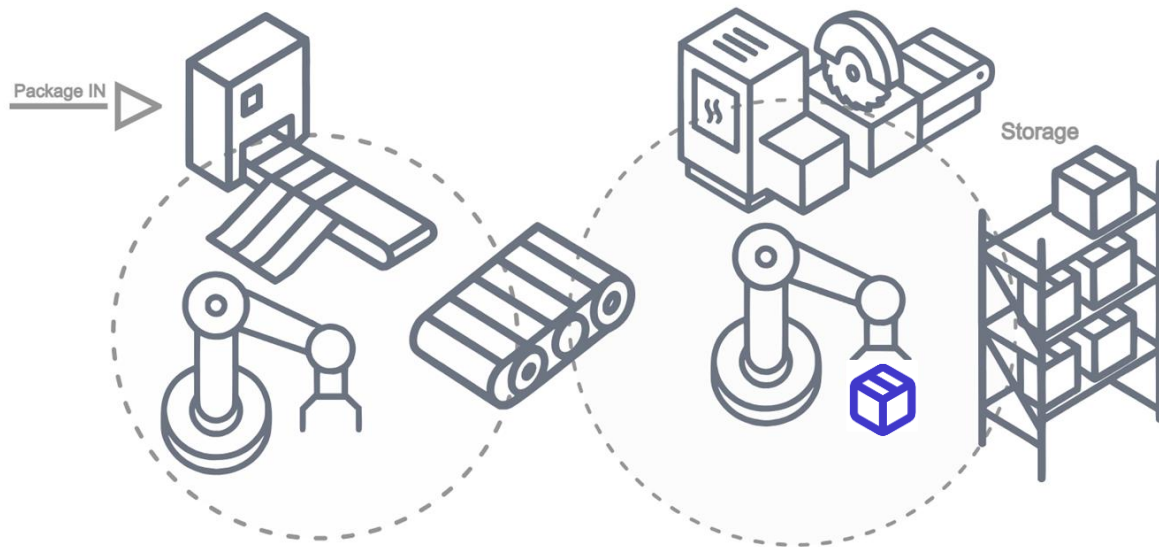
Setup



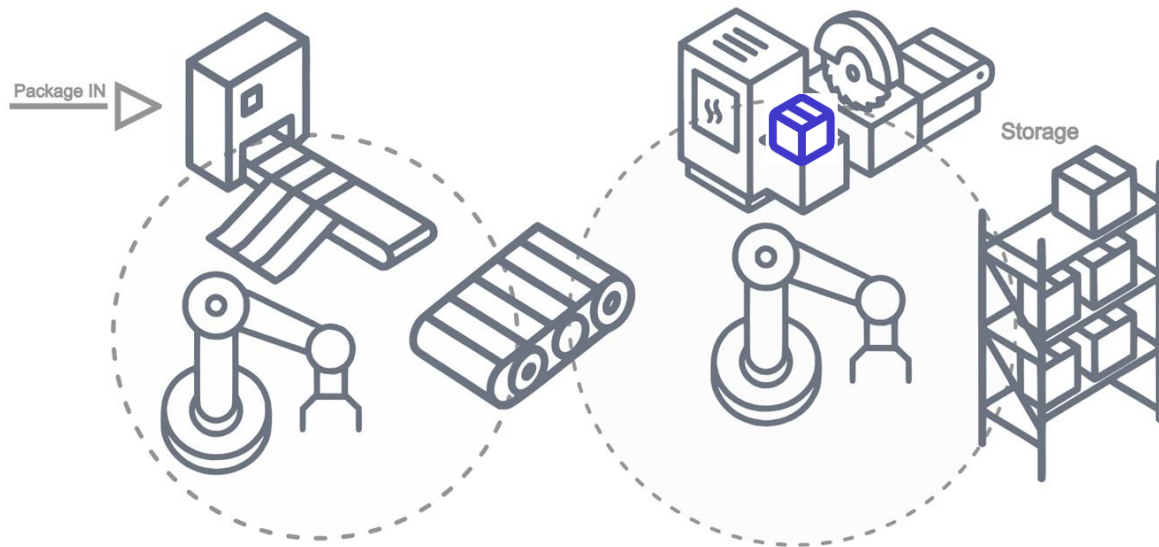
Setup



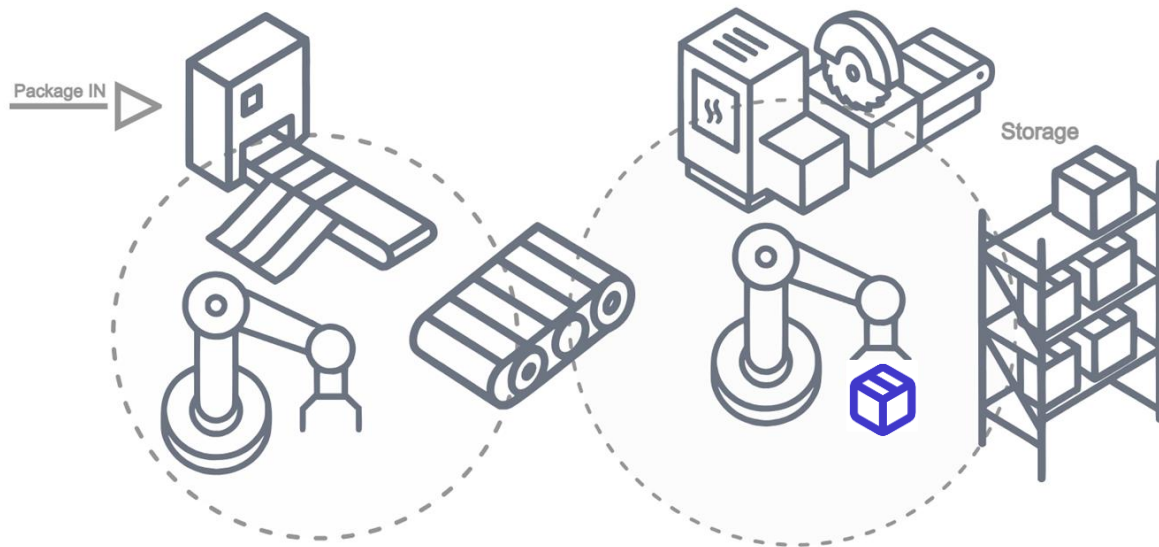
Setup



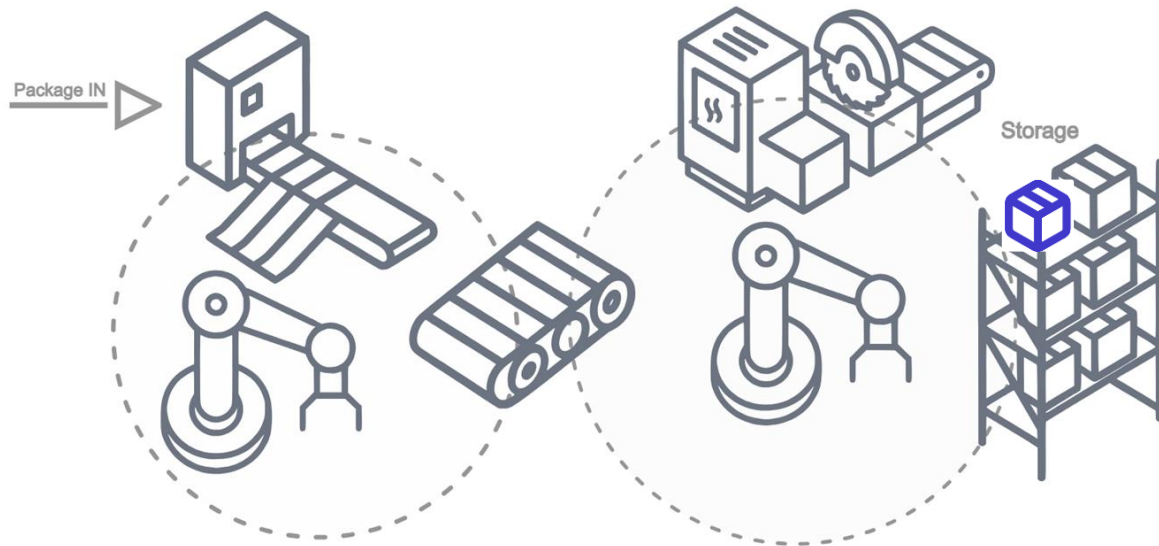
Setup



Setup



Setup





Use cases

- Monitoring the status of the production line
- Simulating various scenarios:
 - E.g., loss of a redundant machine and recalibration of the process, ...
- Analysis
 - E.g., energy usage, errors, throughput,...
- Optimization
 - E.g., energy usage, speed, orders,
- Prediction
 - E.g., when will a part break, should be replaced,...



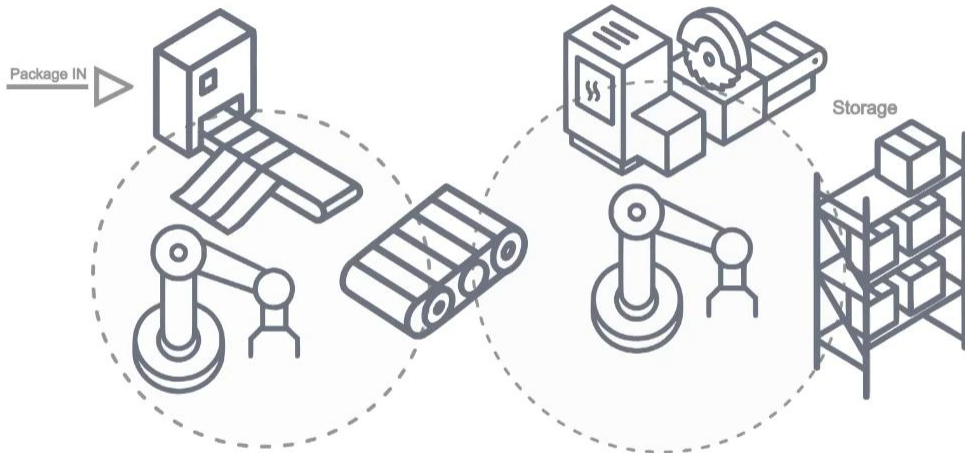
Use cases

- **Monitoring the status of the production line**
- **Simulating various scenarios:**
 - E.g., loss of a redundant machine and recalibration of the process, ...
- **Analysis**
 - E.g., energy usage, errors, throughput,...
- **Optimization**
 - E.g., energy usage, speed, orders,
- **Prediction**
 - E.g., when will a part break, should be replaced,...



Monitoring the production line

Production Flow Visualization



Packages

No packages active

Machine Status Dashboard

ConveyorBelt01

● Idle

Direction: Backward
Conveyor: Stopped

VacuumGripper02

● Idle

Rotation: 42.4°
Height: 54mm
Forward: 124mm

SortingLine01

● Idle

Conveyor: Stopped
Color: RED
Branch: MIDDLE

VacuumGripper01

● Running

Rotation: 61.2°
Height: -1mm
Forward: 0mm

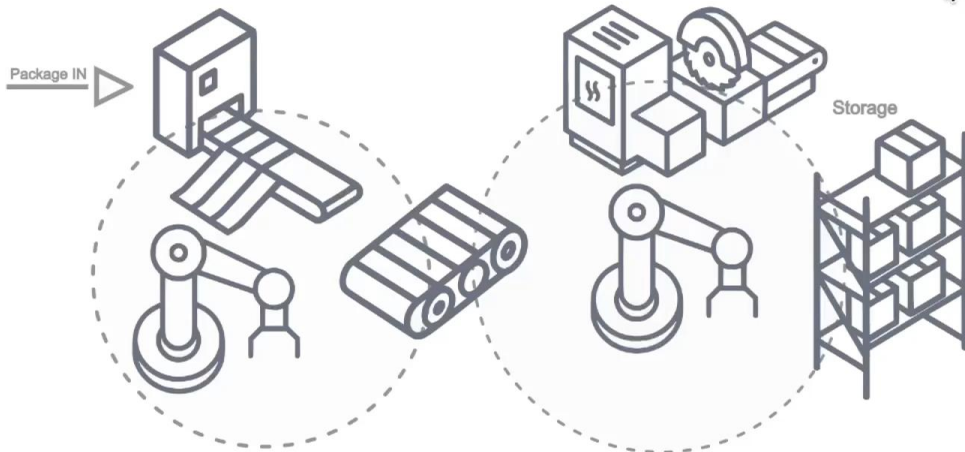


Simulating various scenarios

Simulation based on these parameters

- Simulated time
- Simulation's speedup
- Package generation interval
- Machine-specific speedup

Production Flow Visualization



Packages

No packages active

Machine Status Dashboard

ConveyorBelt01

● Idle

Direction: Backward
Conveyor: Stopped

VacuumGripper02

● Idle

Rotation: 124.7°
Height: 54mm
Forward: 124mm

SortingLine01

● Idle

Conveyor: Stopped
Color: RED
Branch: MIDDLE

VacuumGripper01

● Idle

Rotation: 296.5°
Height: 0mm
Forward: 0mm



Analysis

Simulation based on these parameters

- **Simulated time**
- **Simulation's speedup**
- **Package generation interval**
- **Machine-specific speedup**

Implemented two virtual sensors

- **Heating**
- **Energy usage**

Start Simulation

Stop Simulation

 Current Configuration Save Preset Reset

Sim: 1440min (1d 0h 0m)

Duration: 4m 47s

300x (Extremely Fast)

2.1min/pkg

 Simulation Status

SIMULATION ID

51

NAME

Simulation_2025-09-24T22:03:25.04

STARTED

Sep 24 at 22:03

Progress

10.7%

 Elapsed: 30s

Duration: 4m:48s

 Running 8 Overheating Events

Processed

71

Expected

686

 Basic Parameters

Core simulation timing and behavior settings

Simulation Time to Simulate

Days

Hours

Minutes

Optimization

Simulation based on these parameters

- Simulated time
- Simulation's speedup
- Package generation interval
- Machine specific speedup

Implemented two virtual sensors

- Heating
- Energy usage

optimization goals

 Packages-Per-Hour

Maximize throughput of packages per hour

 Energy-Per-Hour

Minimize total energy consumption per hour

 Energy-Per-Package

Minimize energy used per processed package

 Overheating Events & Issues

Most Overheated Machine

Machine: **ConveyorBelt01, VacuumGripper02, SortingLine01, VacuumGripper01, MultiProcessing01, HighBay01**

Energy-Per-Package:

Wh

Overheating:

0

Highest Energy-Per-Package

Machine: **HighBay01** Energy-Per-Package: 1.83 Wh Overheating: 0

Machine-Specific Overheating Analysis

ConveyorBelt01

0 overheating events

VacuumGripper02

0 overheating events

SortingLine01

0 overheating events

VacuumGripper01

0 overheating events

MultiProcessing01

0 overheating events

HighBay01

0 overheating events

 Optimization Opportunities

Current Machine Power Multipliers

Digital Twin

Technical Architecture



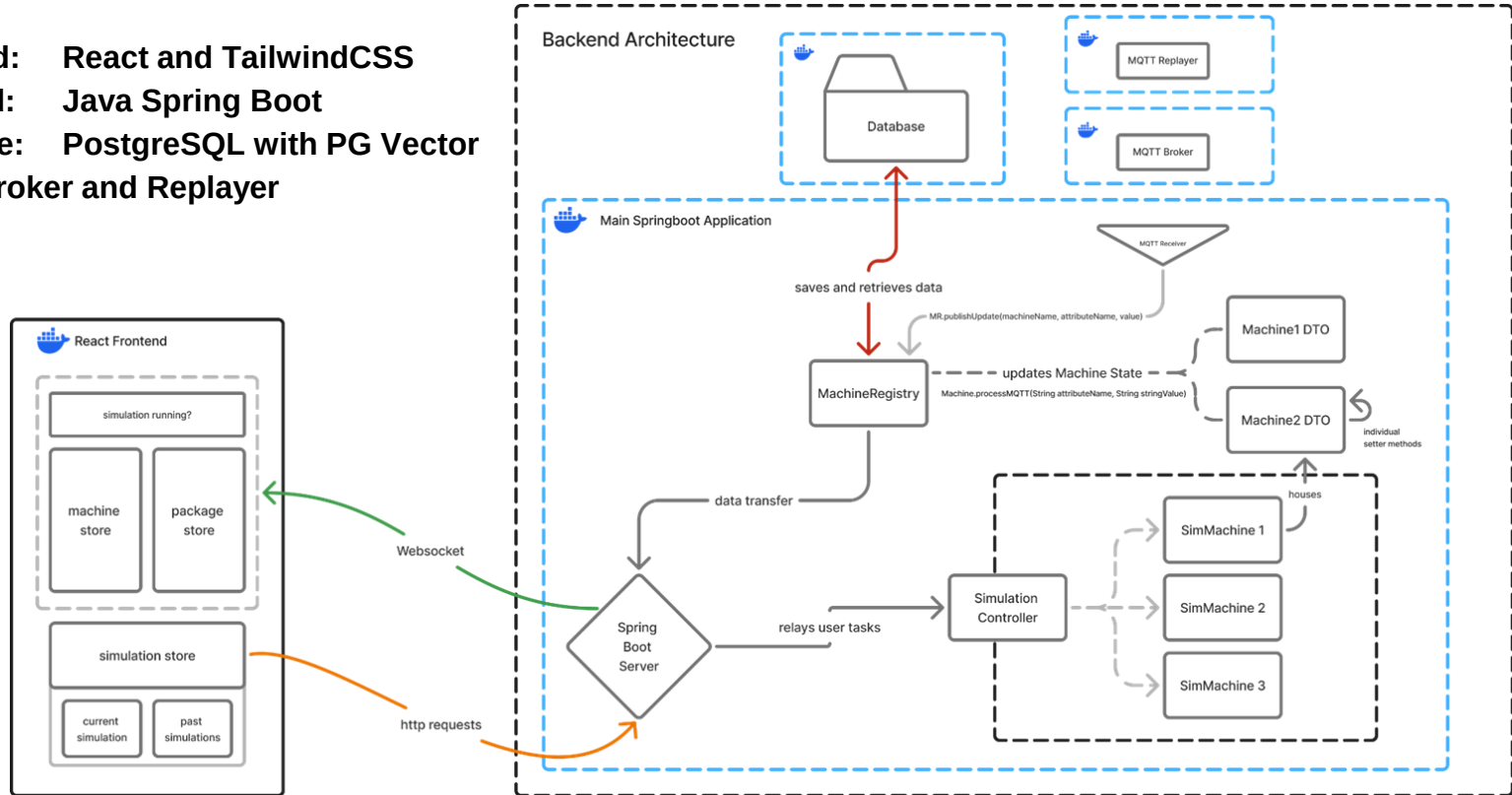
Technical Architecture Sketch

Frontend: React and TailwindCSS

Backend: Java Spring Boot

Database: PostgreSQL with PG Vector

MQTT Broker and Replayer





Optimization

- **Given a set of fixed parameters**
- **Task: optimize the other parameters to improve on one of the following metrics:**
 - **Packages per hour**
 - **Energy per hour**
 - **Energy per package**
- **Generate 1000 simulation parameters randomly (with fixed parameters set)**
- **Predict the simulation result for each with kNN**
- **Return the parameters with best result for chosen metric**

[Start Simulation](#)
[Stop Simulation](#)

Current Configuration

[Save Preset](#)
[Reset](#)

Sim: 100min (0d 1h 40m)

Duration: 12s

500x (Extremely Fast)

1min/pkg

Simulation Status

[Complete](#)
[View Details](#)

✕

 SIMULATION ID
867

 NAME
Simulation_2025-09-25T09:48:58.14

 STARTED
Sep 25 at 09:48 AM

Progress

100.0%

Elapsed: 12s

Duration: 12s

[Complete](#)
[3 Overheating Events](#)

 Processed
52

 Expected
100

Basic Parameters

Core simulation timing and behavior settings

Simulation Time to Simulate

Days

0

Hours

1

Minutes

40

Total: 100 minutes | Real duration: 12s

Simulation Speed

500x (Extremely Fast)

1x (Real-time)

250x

500x (Fast)



Optimization

Why don't we use brute force to find the best simulation parameters?

- **600 Million simulation parameter combinations**
- **Simulation time of 1 hour and speedup of 500 => 180 years**
- **even with 180 cores you would need 1 year**
- **highly flexible architecture => rapid changes possible**



k-Nearest-Neighbor (kNN)

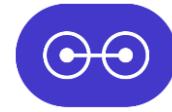
- **7 simulation parameters represented as Vectors in the DB**
(package generation interval, 6 machine specific speedup values)
- **Database contains 20.000 simulation results, which cover the parameter space => consistency of kNN**
- **Implementation of kNN with PG Vector Extension:**

```
55 String sql =  
56     "SELECT sbd.id, sdd.total_packages, sdd.total_energy_usage," +  
57     "    sp.simulation_days, sp.simulation_hours, sp.simulation_minutes "  
58     "FROM simulation_base_data sbd " +  
59     "INNER JOIN simulation_data_details sdd ON sbd.id = sdd.id " +  
60     "INNER JOIN simulation_parameters sp ON sbd.parameters_id = sp.id " +  
61     "WHERE sdd.parameters_vec IS NOT NULL " +  
62     "ORDER BY sdd.parameters_vec <-> ?::vector " +  
63     "LIMIT ?";
```





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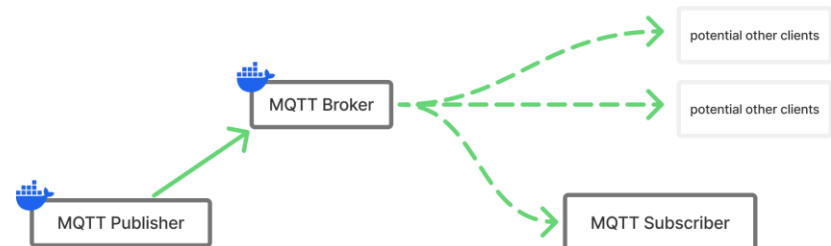


Data Analysis



Starting Point

PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEncoderCounter	{\"value\": 1770, \"timestamp\": \"2025-05-12T11:59:13.992112+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEncoderCounter	{\"value\": 2690, \"timestamp\": \"2025-05-12T11:59:13.992289+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEncoderCounter	{\"value\": 909, \"timestamp\": \"2025-05-12T11:59:13.992372+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEndUp	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.992448+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEnd	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.992523+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEndIn	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.992597+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/internal/isExecuting	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.949683+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalUp	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.950466+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalDown	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.950781+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotRight	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.950989+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotLeft	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.951351+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmOut	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.951749+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmIn	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.952120+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActCompressorOn	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.952287+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActValve	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.952406+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/machine_feedback	{\"value\": \"{\\\"jsonType\\\": \\\"MACHINE_FEEDBACK\\\", \\\"status\\\": \\\"UNINITIA
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/command_feedback	{\"value\": \"{\\\"jsonType\\\": \\\"COMMAND_FEEDBACK\\\", \\\"commandId\\\": 8, \\
PLC/Island 1/VacuumGripper/VacuumGripper01/events/received/command	{\"value\": \"{\\\"COMMAND VACUUM 8 MOVE (meaning START vertical 1300 ro
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEncoderCounter	{\"value\": 807, \"timestamp\": \"2025-05-12T11:59:13.952982+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensRotEncoderCounter	{\"value\": 1782, \"timestamp\": \"2025-05-12T11:59:13.953249+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensArmEncoderCounter	{\"value\": 1865, \"timestamp\": \"2025-05-12T11:59:13.953569+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEndUp	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.953842+00:00\"}
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PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/internal/isExecuting	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.955005+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalUp	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.955005+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalDown	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.955707+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActRotRight	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.955793+00:00\"}
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PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmOut	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.955957+00:00\"}
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmIn	{\"value\": false, \"timestamp\": \"2025-05-12T11:59:13.956029+00:00\"}





Data Structure

PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEncoderCounter	{ "value": 1770, "timestamp": "2025-05-12T11:59:13.992112+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEncoderCounter	{ "value": 2690, "timestamp": "2025-05-12T11:59:13.992289+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEncoderCounter	{ "value": 909, "timestamp": "2025-05-12T11:59:13.992372+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEndUp	{ "value": false, "timestamp": "2025-05-12T11:59:13.992448+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEnd	{ "value": false, "timestamp": "2025-05-12T11:59:13.992523+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEndIn	{ "value": false, "timestamp": "2025-05-12T11:59:13.992597+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/internal/isExecuting	{ "value": false, "timestamp": "2025-05-12T11:59:13.949683+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalUp	{ "value": false, "timestamp": "2025-05-12T11:59:13.950466+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalDown	{ "value": false, "timestamp": "2025-05-12T11:59:13.950781+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotRight	{ "value": false, "timestamp": "2025-05-12T11:59:13.950989+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotLeft	{ "value": false, "timestamp": "2025-05-12T11:59:13.951351+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmOut	{ "value": false, "timestamp": "2025-05-12T11:59:13.951749+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmIn	{ "value": false, "timestamp": "2025-05-12T11:59:13.952120+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActCompressorOn	{ "value": false, "timestamp": "2025-05-12T11:59:13.952287+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActValve	{ "value": false, "timestamp": "2025-05-12T11:59:13.952406+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/machine_feedback	{ "value": "{ \"jsonType\": \"MACHINE_FEEDBACK\", \"status\": \"UNINITIA\" }", "timestamp": "2025-05-12T11:59:13.952406+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/command_feedback	{ "value": "{ \"jsonType\": \"COMMAND_FEEDBACK\", \"commandId\": \"8\" }", "timestamp": "2025-05-12T11:59:13.952406+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper01/events/received/command	{ "value": "{ \"jsonType\": \"COMMAND_VACUUM 8 MOVE (meaning START vertical 1300 r\" }", "timestamp": "2025-05-12T11:59:13.952406+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEncoderCounter	{ "value": 807, "timestamp": "2025-05-12T11:59:13.952982+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensRotEncoderCounter	{ "value": 1782, "timestamp": "2025-05-12T11:59:13.953249+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensArmEncoderCounter	{ "value": 1865, "timestamp": "2025-05-12T11:59:13.953569+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEndUp	{ "value": false, "timestamp": "2025-05-12T11:59:13.953842+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensRotEnd	{ "value": false, "timestamp": "2025-05-12T11:59:13.954118+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensArmEndIn	{ "value": false, "timestamp": "2025-05-12T11:59:13.954380+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/internal/isExecuting	{ "value": false, "timestamp": "2025-05-12T11:59:13.955005+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalUp	{ "value": false, "timestamp": "2025-05-12T11:59:13.955005+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalDown	{ "value": false, "timestamp": "2025-05-12T11:59:13.955707+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActRotRight	{ "value": false, "timestamp": "2025-05-12T11:59:13.955793+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActRotLeft	{ "value": false, "timestamp": "2025-05-12T11:59:13.955882+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmOut	{ "value": false, "timestamp": "2025-05-12T11:59:13.955957+00:00Z" }
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmIn	{ "value": false, "timestamp": "2025-05-12T11:59:13.956029+00:00Z" }

1. Column: MQTT topic string

This column tells us where the message is coming from.

```
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEncoderCounter
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEncoderCounter, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEncoderCounter, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEndUp, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEnd, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEndIn, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalUp, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalDown, "{
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotRight, "{
```

- **PLC/Island 1:** Controller/Production Island (always the same)
- **VacuumGripper:** Device type, asset family (robot arm etc.)
- **VacuumGripper1:** Device instance ID
- **Measurements:** Traffic class (measurements/events/commands)
 - **Measurements:** PT → DT raw values
 - **events:** higher-level events (Cycle finished etc., most often DT → anyone)
 - **commands:** DT → PT control (not necessary due to missing connection to PT → filter out?)
- **Input:** Domain (Input/Output)
- **vacuumSensVerticalEncoderCounter:** Signal name

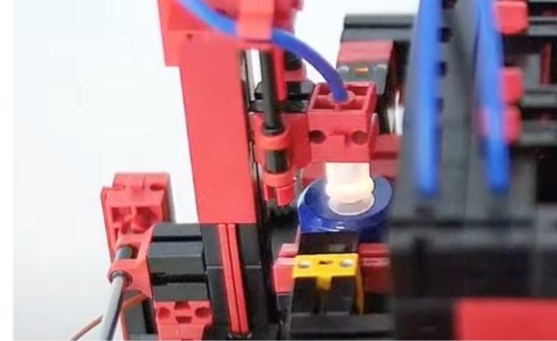
2. Column: the payload JSON

This column tells us about the content of the message.



Data Analysis

PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEncoderCounter	("value": 1770, "timestamp": "2025-05-12T11:59:13.992112+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensRotEncoderCounter	("value": 2690, "timestamp": "2025-05-12T11:59:13.992289+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEncoderCounter	("value": 909, "timestamp": "2025-05-12T11:59:13.992372+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEndUp	("value": false, "timestamp": "2025-05-12T11:59:13.992448+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensVerticalEnd	("value": false, "timestamp": "2025-05-12T11:59:13.992523+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/input/vacuumSensArmEndIn	("value": false, "timestamp": "2025-05-12T11:59:13.992597+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/internal/isExecuting	("value": false, "timestamp": "2025-05-12T11:59:13.949683+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalUp	("value": false, "timestamp": "2025-05-12T11:59:13.950466+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActVerticalDown	("value": false, "timestamp": "2025-05-12T11:59:13.950781+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotRight	("value": false, "timestamp": "2025-05-12T11:59:13.950989+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActRotLeft	("value": false, "timestamp": "2025-05-12T11:59:13.951351+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmOut	("value": false, "timestamp": "2025-05-12T11:59:13.951749+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActArmIn	("value": false, "timestamp": "2025-05-12T11:59:13.952120+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActCompressorOn	("value": false, "timestamp": "2025-05-12T11:59:13.952287+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/measurements/output/vacuumActValve	("value": false, "timestamp": "2025-05-12T11:59:13.952406+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/machine_feedback	("value": {"@jsonType": "MACHINE_FEEDBACK", "status": "UNINITIA
PLC/Island 1/VacuumGripper/VacuumGripper01/events/emitted/command_feedback	("value": {"@jsonType": "COMMAND_FEEDBACK", "commandId": "8",
PLC/Island 1/VacuumGripper/VacuumGripper01/events/received/command	("value": "COMMAND VACUUM 8 MOVE (meaning START vertical 1300 r
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEncoderCounter	("value": 807, "timestamp": "2025-05-12T11:59:13.952982+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensRotEncoderCounter	("value": 1782, "timestamp": "2025-05-12T11:59:13.953249+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensArmEncoderCounter	("value": 1865, "timestamp": "2025-05-12T11:59:13.953569+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEndUp	("value": false, "timestamp": "2025-05-12T11:59:13.953842+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensVerticalEnd	("value": false, "timestamp": "2025-05-12T11:59:13.954118+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/input/vacuumSensArmEndIn	("value": false, "timestamp": "2025-05-12T11:59:13.954380+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/internal/isExecuting	("value": false, "timestamp": "2025-05-12T11:59:13.955005+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalUp	("value": false, "timestamp": "2025-05-12T11:59:13.955505+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActVerticalDown	("value": false, "timestamp": "2025-05-12T11:59:13.955707+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActRotRight	("value": false, "timestamp": "2025-05-12T11:59:13.955793+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActRotLeft	("value": false, "timestamp": "2025-05-12T11:59:13.955882+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmOut	("value": false, "timestamp": "2025-05-12T11:59:13.955957+00:00")
PLC/Island 1/VacuumGripper/VacuumGripper02/measurements/output/vacuumActArmIn	("value": false, "timestamp": "2025-05-12T11:59:13.956029+00:00")



18948 ProcessingMultiProcessing01measurements/output/multiProcessingActRotCounterClockwise	("value": false, "timestamp": "2025-05-12T12:01:37.894970+00:00")
18949 ProcessingMultiProcessing01measurements/output/multiProcessingActConveyerForward	("value": false, "timestamp": "2025-05-12T12:01:37.894970+00:00")
18950 ProcessingMultiProcessing01measurements/output/multiProcessingActSaw	("value": false, "timestamp": "2025-05-12T12:01:37.895258+00:00")
18951 ProcessingMultiProcessing01measurements/output/multiProcessingActOverhead	("value": false, "timestamp": "2025-05-12T12:01:37.895619+00:00")
18952 ProcessingMultiProcessing01measurements/output/multiProcessingActOverOutward	("value": true, "timestamp": "2025-05-12T12:01:37.895943+00:00")
18953 ProcessingMultiProcessing01measurements/output/multiProcessingActGripperToChen	("value": true, "timestamp": "2025-05-12T12:01:37.896282+00:00")
18954 ProcessingMultiProcessing01measurements/output/multiProcessingActGripperToTurntable	("value": false, "timestamp": "2025-05-12T12:01:37.897113+00:00")
18955 ProcessingMultiProcessing01measurements/output/multiProcessingOverLight	("value": false, "timestamp": "2025-05-12T12:01:37.897470+00:00")

- SensTurntablePosVacuum set to "true" -> Gripper arrived at the turntable

13943 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensVacuumGripperActOpen	("value": false, "timestamp": "2025-05-12T12:01:39.103065+00:00")
13944 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensOverFeederIn	("value": false, "timestamp": "2025-05-12T12:01:39.103477+00:00")
13945 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensOverFeederOut	("value": false, "timestamp": "2025-05-12T12:01:39.103834+00:00")
13946 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensTurntablePosVacuum	("value": true, "timestamp": "2025-05-12T12:01:39.104402+00:00")
13947 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensTurntablePosBell	("value": false, "timestamp": "2025-05-12T12:01:39.104700+00:00")
13948 IutlProcessingMultiProcessing01measurements/input/multiProcessingSensTurntablePosSaw	("value": false, "timestamp": "2025-05-12T12:01:39.104937+00:00")

- ActRotClockwise set to "true" -> Rotator turns clockwise and delivers package at saw

14073 PLCIsland 1MultiProcessingMultiProcessing01measurements/output/multiProcessingValveOverDoor	("value": false, "timestamp": "2025-05-12T12:01:55.458265+00:00")
14074 PLCIsland 1MultiProcessingMultiProcessing01measurements/output/multiProcessingValveOverDoor	("value": false, "timestamp": "2025-05-12T12:01:55.458265+00:00")



Data Analysis

Detailed Analysis

For the detailed analysis of the **MQTT data** and its elements, please refer to:

- [Data Structure Analysis](#)

For the analysis of **individual machines**

- [ConveyorBelt](#)
- [HighBay](#)
- [MultiProcessing](#)
- [SortingLine](#)
- [VacuumGripper](#)

ConveyorBelt:

- String machineName
- MachineState state
- LocalDateTime timestamp
- boolean forward
- boolean conveyorRunning

HighBay:

- String machineName
- MachineState state
- LocalDateTime timestamp
- boolean dirty
- String ejectedToBranch

MultiProcessing:

- String machineName
- MachineState state
- LocalDateTime timestamp
- String colorPresent
- boolean conveyorRunning
- boolean dirty
- String ejectedToBranch

SortingLine:

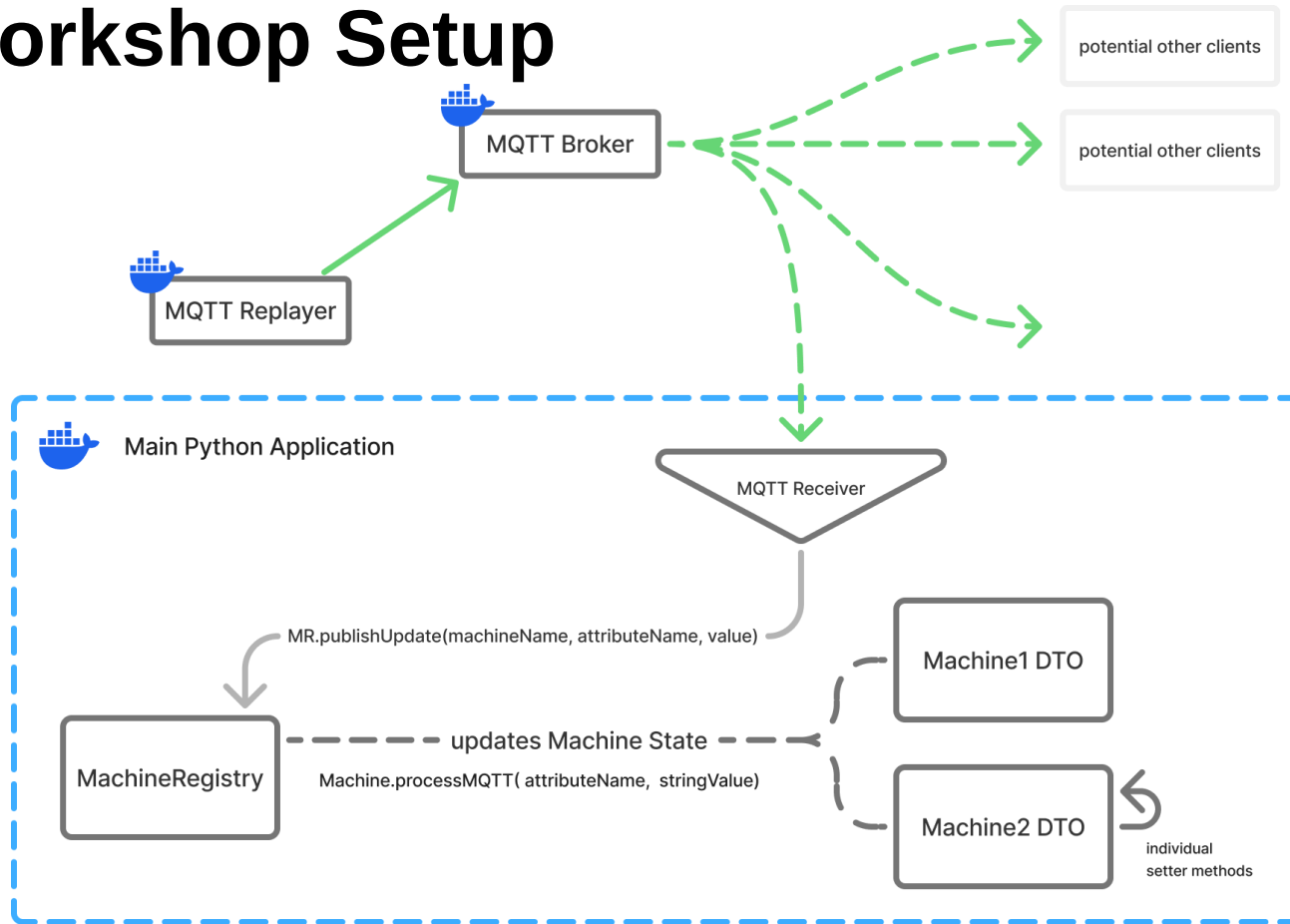
- String machineName
- MachineState state
- LocalDateTime timestamp
- String colorPresent
- boolean conveyorRunning
- boolean dirty
- String ejectedToBranch

VacuumGripper:

- String machineName
- MachineState state
- LocalDateTime timestamp
- boolean dirty
- String ejectedToBranch



Workshop Setup





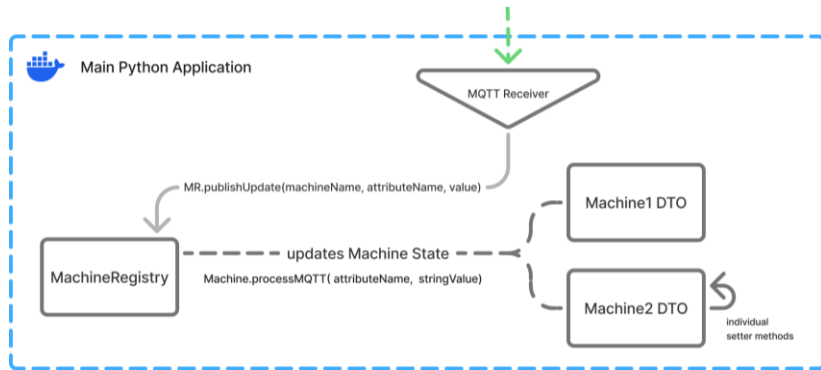
Your Task

- Download the example project from the Repo <https://github.com/sheffdatasoc/2025-26/>
- Run the docker compose to start the MQTT replayer and broker
- Start the workshop_example.py
- Monitor your terminal to see the example running!
- Try implementing the conveyorbelt yourself (the skeleton is given in the conveyor_belt.py)
- Run workshop_example again and see the conveyorbelt change!

ConveyorBelt:

- String machineName
- MachineState state
- LocalDateTime timestamp
- boolean forward
- boolean conveyorRunning

You should implement the marked attributes





Your Task – Step by step

- Look at how the `process_mqtt` is implemented in the other classes
- Print the `attribute_name` and the value the `conveyor_belt` receives from `mqtt`
- Think about how to adapt the of the other classes for the `conveyor_belts` `mqtt` attribute names
- Implement the `to_dict` and `__repr__` function

ConveyorBelt:

- `String machineName`
- `MachineState state`
- `LocalDateTime timestamp`
- `boolean forward`
- `boolean conveyorRunning`

You should implement the marked attributes

